PADHAKU
Exploration · Class 9 Science



CHAPTER 11

Reproduction: How Life Continues

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples; write your own.

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🤔 Think It Over

1 **When does a farmer prefer asexual or sexual methods of reproduction for crops production?**

✔ ANSWER

- **Asexual (vegetative) methods** when the farmer wants many plants quickly that are exactly like a good parent plant (same taste, yield, colour), e.g. sugarcane, potato, banana, rose. Also for plants that make few or no good seeds.
- **Sexual methods (seeds)** when the farmer wants new varieties with new useful characters (high yield, disease resistance) by mixing characters of two parents, or for crops grown easily from seeds like wheat, rice and maize.

2 **Why do you think most complex animals and flowering plants use sexual reproduction, while many simple organisms, like yeast and hydra mainly reproduce asexually?**

✔ ANSWER

- Sexual reproduction mixes characters of two parents and creates **variation**. Variation helps complex organisms, which live long and reproduce slowly, adapt to changing environments.

- Simple organisms like yeast and hydra need to multiply **fast**. Asexual reproduction needs only one parent, is quick and makes many copies when conditions are good.

🧪 Activity 11.1 · 11.1.1 Vegetative propagation in agriculture

Cutting (i) **Does the gardener cut the overgrown branches of a plant at the end of its growing season?**

💡 SAMPLE ANSWER

Yes. Overgrown branches are usually cut (pruned) at the end of the growing season. These healthy branches are used to make cuttings.

Cutting (ii) **Observe them prepare the cuttings. Note the average length of the cuttings.**

💡 SAMPLE ANSWER

Cuttings are usually about **15–20 cm** long, cut from a healthy, mature stem (length differs from plant to plant).

Cutting (iii) **Count the number of nodes and internodes on the cuttings.**

💡 SAMPLE ANSWER

A cutting usually has **3–4 nodes** and the internodes between them. Roots grow from the nodes buried in the soil and new shoots grow from the buds at the upper nodes.

Observation

Cutting, grafting and layering: water the plants regularly and observe the change, if any.

✔ ANSWER

- **Cutting:** after some days, roots form at the lower nodes and new leaves appear; each cutting grows into a new plant.
- **Grafting:** the stem of Plant B joins with Plant A and grows; it gives flowers of Plant B (e.g. yellow/pink rose) while using the roots of Plant A.
- **Layering:** after 10–15 days roots develop on the buried part of the twig; when cut from the parent, it grows as a new plant.

Conclusion: all these methods use only one parent, so the new plants are identical to the parent (asexual reproduction).

Activity 11.2 · Budding in yeast

? **Do you observe any small, round outgrowths (buds) emerging from the parent yeast cells? Do these features indicate that the yeast is duplicating? How do these observations help you in understanding reproduction in yeast?**

✔ ANSWER

- Yes, small round buds are seen growing out of many yeast cells; some buds may have their own buds, forming a chain.
- Yes, it shows the yeast cell is making a copy of itself.
- So yeast reproduces **asexually by budding**: a bud grows on the parent cell, enlarges and finally separates to live as a new yeast cell.

Activity 11.3 · Spore formation in bread mould

5 **After three days, observe the surface of the bread or roti using a magnifying glass. Do you notice the growth of mould?**

✔ ANSWER

Yes. White cottony threads appear first, which later show tiny black (or greenish) dots. This is mould (a fungus).

8–9 **Observe the mould under the microscope, draw its diagram and compare it with Fig.**

11.8.

✔ ANSWER

We see thin, thread-like **hyphae** with round **sacs** at their tips (like Rhizopus, Fig. 11.8a). The sacs contain many tiny round **spores**. Draw the hyphae, the sac and the spores as in Fig. 11.8a.

? **Do you observe thread-like structures with a round sac at the tip? Do you also see tiny round structures (spores) inside the sac? Where did the mould on your bread slice come from?**

✔ ANSWER

- Yes, thread-like hyphae with round sacs at the tips are seen.
- Yes, the sacs contain many tiny round spores.
- The mould came from **spores present in the air**. They settled on the moist bread, and with warmth and moisture they germinated and multiplied quickly.

Bridging Science and Society **Do you know any fungus that can degrade plastic?**

💡 SAMPLE ANSWER

Yes. Some fungi can break down plastics, e.g. **Aspergillus tubingensis** and **Pestalotiopsis microspora** can degrade polyurethane plastic.

? **How many combinations of characters can gametes carry?**

✔ ANSWER

Very many. During meiosis, each gamete gets one chromosome from each pair at random, so gametes carry many different combinations of characters (see Activity 11.4).

4 **How many combinations can you make with just these three pairs of characters (beads)?**

✔ ANSWER

Each pair gives 2 choices.

Combinations = $2 \times 2 \times 2 = 2^3 =$ **8**

Pair 1	Pair 2	Pair 3
light green	light blue	light red
light green	light blue	dark red
light green	dark blue	light red
light green	dark blue	dark red
dark green	light blue	light red
dark green	light blue	dark red
dark green	dark blue	light red
dark green	dark blue	dark red

5 **Imagine how many combinations are possible with 23 pairs of chromosomes.**

✔ ANSWER

Each of the 23 pairs gives 2 choices.

Combinations = 2^{23} = **83,88,608 (about 84 lakh)** for one gamete.

That is why children differ from their parents and from their siblings.

11.2.2 Sexual reproduction in flowering plants · Activity 11.5

? **Flowers are coloured and/or fragrant. How are these features of flowers useful for reproduction?**

✔ ANSWER

Bright colours, fragrance and nectar **attract insects and birds** (pollinators). They carry pollen from one flower to another and help in pollination, which is needed for forming fruits and seeds.

? **Do you observe thin, flat, green covering and coloured projections present in both (flowers and buds)?**

✔ ANSWER

Yes. Both have the green covering (sepals) and the coloured parts (petals). In the bud, the petals are covered by the sepals.

? **Do you think that sepals protect flowers in the bud stage, along with other parts of the flower when it blooms?**

✔ ANSWER

Yes. Sepals cover and protect the inner parts of the flower in the bud stage. After blooming they support the petals and other parts from below.

? **Do all flowers have these two parts? What are the other parts of a flower?**

✔ ANSWER

No, not all flowers have both; in some flowers sepals and petals are fused, and some flowers (e.g. grasses like wheat) have no showy petals.

Other parts: **stamens** (male part: filament + anther) and **pistil** (female part: stigma, style, ovary).

3–6 **Record the presence of floral parts in different flowers in Table 11.1 and guess the function of each part.**

💡 SAMPLE ANSWER

Flower part	Hibiscus	Mustard	Function (guess)
Sepal	Yes	Yes	Protects the bud
Petal	Yes	Yes	Attracts pollinators
Stamen	Yes	Yes	Makes pollen (male gametes)
Pistil	Yes	Yes	Has ovules; forms fruit and seeds

Other features: the ovary (swollen base of pistil) shows small round **ovules** inside when cut in T.S. and L.S. Draw what you see.

? **Can you guess the function of the most attractive part of a flower — the coloured petals?**

✔ ANSWER

Coloured petals **attract insects and birds** for pollination. They also protect the stamens and pistil.

Activity 11.6 · Pollination in a pea plant

8 **Note your observations in Table 11.2. In which treatment(s) do you find the flowers are replaced by fruits? What can we infer from this activity?**

✔ ANSWER

Treatment	Fruit formation
Flower bud (wrapped)	Yes
Flower bud with removed stamens (wrapped)	No
Flower with removed stamens (wrapped)	Yes
Flower (wrapped)	Yes
Flower (without bag)	Yes

Fruits form in all treatments **except the bud whose stamens were removed** (no pollen ever reached its stigma). A blossomed flower had already received pollen before its stamens were removed.

Inference: transfer of pollen grains from anther to stigma (**pollination**) is necessary for fruit formation.

? What do you think will happen if pollination does not occur?

✔ ANSWER

Without pollination, male gametes cannot reach the egg, so **fertilisation will not happen**. No seeds and fruits will form; the flower will usually wither and fall off.

Activity 11.7 · 11.2.5 Pollination strategies and seed formation

? **What is the approximate success rate of the common strategies — wind pollination and insect pollination — with respect to the seed formation?**

✔ ANSWER

Wind pollination has a **very low** success rate (only about 1 seed for every 5,000–10,000 pollen grains). Insect pollination has a **much higher** success rate (about 1 seed for every 20–50 pollen grains). See Activity 11.7.

1 **Compare and analyse the two strategies (Table 11.3) in terms of: pollen to seed ratio; efficiency of pollination and seed formation.**

Strategy	Pollen per flower	Seeds formed
Wind (maize, wheat)	5,00,000–10,00,000	50–200
Insect (sunflower)	20,000–40,000	800–1,000

✔ ANSWER

Pollen to seed ratio

Wind: 5,00,000 ÷ 50 to 10,00,000 ÷ 200 → about **5,000–10,000 pollen per seed**.

Insect: 20,000 ÷ 800 to 40,000 ÷ 1,000 → about **25–40 pollen per seed**.

Efficiency: insect pollination is far more efficient — insects carry pollen directly to the sticky stigma of another flower. In wind pollination most pollen is lost in the air, so efficiency is very low.

2 **Explain why producing a very large number of pollen grains can still be an effective pollination strategy.**

✔ ANSWER

- Wind carries pollen randomly, so most of it is wasted.
- By making lakhs of light, small pollen grains, the plant makes sure that at least some land on the long, feathery stigmas of other flowers.
- The plant does not spend energy on colourful petals, nectar or scent, and does not depend on insects. So in open fields with many plants close together (wheat, maize, rice) it works well.

⭐ Pause and Ponder · 11.2 Sexual reproduction in plants

1 **In a china-rose (hibiscus) plant, a pollen tube grows and continues through the style after pollen lands on the stigma. Which process is about to happen next?**

✔ ANSWER

Fertilisation. The male gamete travels through the pollen tube to the ovule and fuses with the egg cell to form a zygote.

2 **Look at the pictures (Fig. 11.16) of calotropis (madar) seeds and dandelion seeds. Can you guess what kind of seed dispersal these seeds are adapted for?**

✔ ANSWER

Dispersal by wind. These seeds are light and have fine silky hairs (like a parachute), so they float in the air and are carried far away by wind.

3 **A farmer plants two varieties of maize side by side, but seeds form only when pollen from one variety reaches the stigma of the other. What type of pollination is this?**

✔ ANSWER

Cross-pollination — pollen goes from a flower of one plant to the stigma of a flower of another plant of the same kind.

Pause and Ponder · 11.3 Sexual reproduction in animals

4 **Why do animals with external fertilisation generally produce more eggs than animals with internal fertilisation?**

✔ ANSWER

In external fertilisation, eggs and sperm are released in water. Many eggs are not fertilised, and many are washed away by water currents or eaten by other animals. Producing a very large number of eggs makes sure that at least some survive.

5 **In animals, which fertilisation method the gametes are more protected?**

✔ ANSWER

In **internal fertilisation**. The gametes meet inside the female body, and the fertilised egg or embryo is protected there, so the young ones have a higher chance of survival.

Threads of Curiosity · What determines a baby's biological sex?

? **Females have XX and males have XY chromosomes. The mother always gives an X; the father gives either an X or a Y. Can you predict who determines the sex of a baby?**

✔ ANSWER

The **father**.

- Egg (X) + sperm with X → XX → **girl**
- Egg (X) + sperm with Y → XY → **boy**

Since only the father's sperm can carry X or Y, the sperm decides the sex of the baby. The mother is not responsible for it.

⭐ Pause and Ponder · 11.5 Reproduction in human beings

6 **Ravi suddenly notices that he is growing taller rapidly, his shoulders are broadening, and his voice cracks. What stage of life is he entering?**

✔ ANSWER

He is entering **puberty** (adolescence), when the reproductive organs mature. These changes are caused by hormones made by the testes.

7 **Rina's period occurs every 28 days. Her last period was on the 5th of March. On which day is she most likely to get her next period?**

✔ ANSWER

Given: cycle = 28 days, last period started 5 March.

March has 31 days: 5 March + 28 days → 31 − 5 = 26 days left in March; 28 − 26 = 2.

Next period: **2nd April**.

8 **A human zygote has just formed. How many chromosomes does it have?**

✔ ANSWER

46 chromosomes (23 pairs): 23 from the sperm + 23 from the egg.

9 **What protective devices can be used during sexual activity to reduce the spread of STIs?**

✔ ANSWER

Condoms (barrier method). They stop direct contact with body fluids, so they reduce the spread of STIs like HIV, gonorrhoea and syphilis, and also prevent pregnancy.

10 **If a couple uses oral contraceptive pills but not condoms, which risks remain and why?**

✔ ANSWER

Pills only change the release of eggs by altering hormones, so they prevent pregnancy. They do **not** stop contact with body fluids, so the risk of **sexually transmitted infections** (HIV, herpes, syphilis, gonorrhoea, etc.) remains. Pills may also have some side effects.

11 **In many animals, the young ones can walk or find food soon after birth but human babies are completely dependent on adults for a**

long time. What might be some advantages and disadvantages of this for humans as a species?

💡 SAMPLE ANSWER

Advantages

- The long childhood gives time for the brain to grow and to learn language, skills and social behaviour from parents and society.
- Strong bonds form in families, and knowledge passes from one generation to the next.

Disadvantages

- Babies are helpless for a long time and need constant care, food and protection.
- Parents spend a lot of time and resources, so humans have few children and the population grows slowly.

⭐ Revise, Reflect, Refine

1 **A flower's anthers are removed before it matures. Later, pollen from another plant of the same species is dusted onto its stigma and seeds are produced. Which process has been ensured here?**
(i) Self-pollination (ii) Cross-pollination (iii) Fertilisation (iv) Tissue culture

✔ ANSWER

(ii) Cross-pollination. The anthers were removed so self-pollination cannot happen; the pollen came from another plant of the same species.

2 **Arrange the following stages of sexual reproduction in plants in the correct order:**
(i) Pollen germination on stigma (ii) Fertilisation (iii) Pollination (iv) Formation of zygote

✔ ANSWER

(iii) Pollination → (i) Pollen germination on stigma → (ii) Fertilisation → (iv) Formation of zygote

3 **Assertion (A): The zygote formed after fertilisation immediately attaches to the uterus wall.**
Reason (R): The uterus wall is always prepared to receive the zygote.
(i) Both A and R are true, and R is the correct explanation of A.
(ii) Both A and R are true, but R is not the correct explanation of A.
(iii) A is true, but R is false.
(iv) A is false, but R is true.

💡 SAMPLE ANSWER

(iv) (the closest option).

A is false: the zygote is formed in the oviduct and goes through many mitotic divisions while travelling to the uterus; only then it implants.

About R: the uterus lining becomes thick and rich in blood vessels every month to receive a zygote. (Strictly, it is prepared in each cycle, not 'always' — it sheds during menstruation.)

4 **Why does asexual reproduction produce offsprings that are genetically identical to the parent?**

✔ ANSWER

Asexual reproduction involves **only one parent** and takes place by **mitosis**. Mitosis makes daughter cells with the same number of chromosomes, identical to the parent cell. There is no mixing of genetic material from two parents, so the offspring are clones of the parent.

5 **Explain why the menstrual cycle stops during pregnancy.**

✔ ANSWER

During pregnancy, the zygote/embryo is implanted in the thick inner lining of the uterus and gets nourishment from it. This lining is needed for the whole pregnancy, so it is **not shed** — hence no menstruation. Also, hormones during pregnancy stop the release of new eggs (no ovulation), so the cycle does not repeat.

6 **Why are flowers that bloom at night white or light in colour as compared to flowers that bloom during the day?**

✔ ANSWER

At night there is little light, so bright colours cannot be seen. White or light-coloured flowers are easily **visible in dim light** (moonlight) to night pollinators like moths and bats. Such flowers are usually also strongly fragrant to attract them.

7 **Why do vegetatively propagated plants tend to be more vulnerable to diseases than sexually reproduced plants?**

✔ ANSWER

Vegetatively propagated plants are **genetically identical** (clones) and have no variation. If a disease attacks one plant, all plants are equally weak against it, so the whole crop can be destroyed. Diseases (e.g. viruses) of the parent also pass into the new plants. Sexually reproduced plants show variation, so some may resist the disease and survive.

8 **If all flowers in a type of plant were only capable of self-pollination, how would it affect the genetic diversity over several generations? Explain.**

✔ ANSWER

Genetic diversity would **decrease**. Pollen and egg come from the same plant, so there is no mixing of characters from different plants. Over many generations the plants would become more and more alike. With little variation, the species would find it hard to adapt to new diseases or changes in the environment.

9 **A farmer wants to produce a large number of genetically identical plants quickly. Suggest suitable reproduction methods and explain why they are effective.**

✔ ANSWER

- **Tissue culture:** many plantlets are made from a small piece (shoot tip) in a lab, quickly and free of viruses (e.g. banana).
- **Cutting, layering, grafting** and planting stem tubers/rhizomes (potato, ginger, sugarcane).

These are asexual methods: only one parent, cell division by mitosis, so all plants are identical to the good parent. They are fast and do not need flowers or seeds.

10 (i) **Suresh prepares slides with pollen grains in different sugar concentrations (0%, 2.5%, 5%, 7.5%, 10%) to study the germination of pollen. What are the different hypotheses which can be tested using this set-up?**

💡 SAMPLE ANSWER

- Pollen grains need sugar to germinate (no germination at 0%).
- Pollen germination increases as the sugar concentration increases.
- There is a best (optimum) sugar concentration for germination; above it, germination decreases.
- Sugar concentration affects the length of the pollen tube.

10 (ii) **What parameters should be kept the same in this set-up?**

✔ ANSWER

Same type (species) and age of pollen; same amount of pollen and of solution on each slide; same temperature, light and humidity; same time of observation; same microscope and method of counting.

11 **Which type(s) of pollination might have been followed in these flowers?**

Tomato	Wheat	Papaya
Stamens cover the stigma.	Flowers open after pollination.	Male and female flowers are often borne on different papaya trees.

✔ ANSWER

- **Tomato: self-pollination.** The stamens cover the stigma, so pollen of the same flower falls on it easily.
- **Wheat: self-pollination.** Pollination is over before the flower opens, so pollen from other plants cannot reach the stigma.
- **Papaya: cross-pollination.** Male and female flowers are on different trees, so pollen must be carried from a male tree to a female tree (by insects or wind).

12 **Apple orchards: Place A — natural pollination; Place B — with beekeeping (bee colony). Fig. 11.24 shows fruit setting and fruit drop:**

	Fruit set (%)	Fruit drop (%)
Place A (natural pollination)	about 26	about 35
Place B (with bee colony)	about 40	about 8

(i) What are the hypotheses? (ii) What are the different parameters? (iii) Compare and analyse the data of Places A and B in terms of high yields. (iv) What do you infer from the data?

✔ ANSWER

(i) Hypotheses

- Keeping bee colonies in apple orchards increases pollination and fruit setting.
- Better pollination by bees reduces fruit drop and increases apple yield.
- The fall in apple yield is due to the decline of natural pollinators.

(ii) Parameters

- Changed (independent): type of pollination — natural pollinators vs beekeeping.
- Measured (dependent): fruit set (%) and fruit drop (%), i.e. yield.
- Kept same: apple variety, age of trees, climate, water, manure, care of orchards.

(iii) Comparison

- Fruit set: B (≈40%) is higher than A (≈26%) — about 14% more.

- Fruit drop: B (≈8%) is much lower than A (≈35%).
- So Place B gives more fruits and keeps most of them → higher yield.

(iv) Inference: beekeeping improves pollination of apple flowers, increasing fruit setting and reducing fruit drop. Mixed farming with bees can raise apple yield and also give honey as extra income.

13 **A student claims, "In humans, ovulation always happens on day 14 of the menstrual cycle". Critically examine this claim and state whether the claim is correct or not. Give at least two reasons.**

✔ ANSWER

The claim is **not correct**.

- The menstrual cycle is not the same for everyone: it is typically 21–35 days long. Day 14 is only for a typical 28-day cycle.
- Ovulation happens about 14 days **before the next period**, so in a 21-day cycle it is around day 7 and in a 35-day cycle around day 21.
- The cycle can change in the same person due to stress, illness, diet, or in the early years after puberty, so ovulation day can shift.

⭐ The Journey Beyond

1 **Read about the 'Seed Village Programme' (Beej Gram Yojana). Why is it important to save indigenous seeds?**

💡 SAMPLE ANSWER

Under the Beej Gram Yojana, farmers are trained to produce and save good-quality seeds in their own village.

Indigenous seeds are important because they are adapted to local soil and climate, resist local pests and droughts, keep genetic diversity alive, can be saved and re-sown every year (lower cost), and are needed to breed new varieties in future.

2 **Prepare a report on IVF, its uses and drawbacks. Discuss it with your class.**

💡 SAMPLE ANSWER

IVF: egg and sperm are joined in a lab dish; the fertilised egg is placed in the uterus.

Uses: helps couples who cannot have children (blocked fallopian tubes, low sperm count, etc.).

Drawbacks: costly; may need many attempts; hormone medicines can have side effects; higher chance of twins/triplets; emotional stress.

3 **Survey crop fields and interact with farmers: crops grown through vegetative propagation and by seeds; indigenous or hybrid seeds; indigenous and hybrid breeds of farm animals; report on how asexual and sexual reproduction are useful in agriculture.**

💡 **SAMPLE ANSWER**

- Vegetative propagation: sugarcane, potato, banana, ginger, turmeric, rose, mango (grafting).
- Seeds: wheat, rice, maize, pulses, mustard, vegetables.
- Many farmers use hybrid seeds for high yield, made by cross-pollination of two selected varieties; some keep indigenous seeds as they are cheaper and suit the local climate.
- Animals: indigenous cows like Gir, Sahiwal, Red Sindhi (hardy, need less care) vs crossbred/hybrid cows like Holstein Friesian or Jersey crosses (more milk, need more care).
- Asexual methods keep good varieties unchanged and fast; sexual methods create new improved varieties.

4 **Explore crop fields near your school and find out pollination strategies: pollinators of different crops; reasons for the declining population of pollinators and their solutions.**

Record pollinators visiting your school garden.

💡 SAMPLE ANSWER

- Pollinators: wind for wheat, rice, maize; bees for mustard, sunflower, pumpkin, brinjal; butterflies and birds for some flowers.
- Reasons for decline: overuse of pesticides, loss of flowering plants and habitats, climate change, diseases of bees.
- Solutions: less and safer use of pesticides, growing flowering plants and hedges, keeping bee boxes (beekeeping), organic farming.

5 **While cutting fruits like lady finger, tomato, brinjal, capsicum, papaya, orange and muskmelon, observe the L.S. and T.S. and attachment of seeds; link them with the structure of the ovary.**

💡 SAMPLE ANSWER

In L.S. and T.S. the fruit shows chambers with seeds attached to its inner wall or centre. The fruit is the ripened **ovary** and the seeds are the **ovules**; the chambers and seed attachment match the inside of the ovary of the flower (e.g. tomato has 2 or more chambers with seeds on a fleshy central part).

⭐ The Quest Continues ...

? Does a unicellular organism like amoeba or yeast ever 'grow old'? When it divides, it produces almost two identical copies. So, does aging happen at all?

💡 **SAMPLE ANSWER**

In amoeba, the parent divides into two daughter cells and no parent body is left to grow old or die, so it is often said to be 'immortal'. But yeast reproducing by budding shows aging: a mother cell can produce only a limited number of buds and then dies. So aging can happen even in some unicellular organisms.

🎉 **Chapter 11 done! Revise the answers once more tomorrow.**